

Summary

In this practical, I learned and implemented two searching techniques: Linear Search and Binary Search. Linear Search checks each element one by one until the required element is found or the list ends, making it simple and suitable for small or unsorted datasets. Binary Search repeatedly divides the sorted array into two halves to locate the target element, making it much faster for large sorted datasets. By performing both algorithms, I understood their working principles, advantages, limitations, and the importance of choosing the appropriate searching method based on the type of data.

Conclusion

From this practical, I concluded that both searching algorithms are useful in different situations. Linear Search is easy to implement and works on both sorted and unsorted data, but it becomes slower as the dataset grows. Binary Search is much more efficient for large datasets because it reduces the search space by half in each step, but it requires the data to be sorted. Overall, this practical improved my understanding of searching techniques and helped me realize how selecting the right algorithm can improve the efficiency and performance of a program.